10 A triangle has vertices A $(1, 4)$, B $(7, 0)$ and C $(-4, -1)$.

(a) Show that the equation of the line AC is $y = x + 3$. **[2]**

M is the midpoint of AB. The line AC intersects the x-axis at D.

(b) Determine the angle DMA. **[7]**

10(a)

10(b)

(answer space continued on next page)

10(b) **(continued)**